

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**ORGANISATION OF ISLAMIC COOPERATION (OIC)****Department of Computer Science and Engineering (CSE)****MID-SEMESTER EXAMINATION****Summer Semester, 2020-2021****DURATION: 1 HOUR 30 MINUTES****FULL MARKS: 75****Math 4441: Probability and Statistics****Programmable calculators are not allowed. Do not write anything on the question paper.**Answer **all 3 (three)** questions. Marks of each question and corresponding CO and PO are written in the right margin.

1.	Suppose there are five coins in an urn. The coins are not same and have different probability of getting a head or tail when any of the coins are tossed. Let the probability of getting a head is p_i when the i th coin is tossed ($i = 1, 2, \dots, 5$). Suppose the value of p_i is given by $p_i = 0.25 \times (i - 1)$, for $i = 1, 2, \dots, 5$. Suppose that you pick a coin randomly from the urn and toss it once.	
a)	Find the probability that the i th coin was picked, π_i , where $i = 1, 2, \dots, 5$, given that a head was obtained when the coin was tossed first time.	10 (CO1) (PO1)
b)	Now assume that you toss for the second time. Find the probability of getting a head again. [Hints: it is the conditional sum rule that calculate the probability of the obtaining a head in the second toss of a randomly selected coin where the first toss also produced a head.]	8 (CO1) (PO1)
c)	If a tail were obtained in the first toss and the same coin were tossed again, find the probability of getting a head in the second toss.	7 (CO1) (PO1)
2.	a) Consider a community of 100 peoples and assume that they have a certain disease. To identify whether any individual of that community has the disease, their blood need to be tested. However, instead of testing everyone separately, it has been decided first to group the people in groups of 10. The blood samples of the 10 people in each group will be pooled and analyzed together. If the test is negative, one test will suffice for the 10 people. If the test is positive, each of the 10 people will also be individually tested. Suppose the probability that a person has the disease is 0.10 for all people independently from each other. Let X represent the number of groups with at least one people having the disease, and therefore, all the 10 people of those groups are individually tested. Find the probability mass function of X .	10 (CO2) (PO2)
b)	A node in a computer network sends data packet to another node one after another. Each packet is successfully sent with probability 0.75 and is dropped with probability 0.25. What is the probability that 5 packets are dropped before 10 packets are successfully sent?	8 (CO2) (PO2)
c)	Suppose that two players (A and B) play a series of games that ends when one of them has won i games. Suppose that each game played is, independently, won by player A with probability p . Suppose N represents the number of games that are played when $i = 3$. Find the PMF of N .	7 (CO2) (PO2)

3.	a)	Let us consider that b is a random number uniformly distributed between -3 and $+3$. Find the probability that the quadratic equation $x^2 + bx + 1 = 0$ has at least one real root.	8 (CO2) (PO2)
	b)	Suppose you roll a four-sided die and X represents the number of dots on the side facing toward you. Now, you toss a coin X time(s), let Y denote the number of tails appears in X tosses. Find the join PMF of X and Y and the marginal PMF of X .	10 (CO2) (PO2)
	c)	The probability density function of X is given by	7 (CO1) (PO2)
		$f_X(x) = \begin{cases} Cxe^{-\frac{x}{2}}, & x > 0 \\ 0, & x \leq 0 \end{cases}$	
		Find the value of C and the probability that $X \geq 5$.	

PMF/PDF, expected value and variance of some known Random Variables

Distribution	PMF/PDF	Expected value	Variance
Bernoulli	$P_X(x) = \begin{cases} 1-p & x=0 \\ p & x=1 \\ 0 & \text{otherwise} \end{cases}$	$E[X] = p$	$Var[X] = p(1-p)$
Geometric	$P_X(x) = \begin{cases} p(1-p)^{x-1} & x \geq 1 \\ 0 & \text{otherwise} \end{cases}$	$E[X] = 1/p$	$Var[X] = (1-p)/p^2$
Binomial	$P_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x = 1, \dots, n \\ 0 & \text{otherwise} \end{cases}$	$E[X] = np$	$Var[X] = np(1-p)$
Pascal	$P_X(x) = \begin{cases} \binom{x-1}{k-1} p^k (1-p)^{x-k} & x = k, k+1, \dots \\ 0 & \text{otherwise} \end{cases}$	$E[X] = k/p$	$Var[X] = k(1-p)/p^2$
Poisson	$P_X(x) = \begin{cases} \frac{(\lambda T)^x e^{-(\lambda T)}}{x!} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$E[X] = \alpha$ $\alpha = \lambda T$	$Var[X] = \alpha$
Hyper Geometric	$P_X(x) = \frac{\binom{r}{x} \binom{g}{n-x}}{\binom{r+g}{n}}$	$E[X] = \frac{rn}{r+g}$	
Uniform (discrete)	$P_X(x) = \begin{cases} \frac{1}{b-a+1}, & x = a, a+1, a+2, \dots, b \\ 0, & \text{otherwise} \end{cases}$	$E[X] = \frac{a+b}{2}$	$Var[X] = \frac{(b-a)(b-a+1)}{12}$
Exponential	$f_X(x) = \begin{cases} ae^{-ax} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$E[X] = 1/a$	$Var[X] = 1/a^2$
Gaussian	$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} & \sigma > 0 \\ 0 & \text{otherwise} \end{cases}$	$E[X] = \mu$	$Var[X] = \sigma^2$
Uniform (Continuous)	$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x < b \\ 0, & \text{otherwise} \end{cases}$	$E[X] = \frac{a+b}{2}$	$Var[X] = \frac{(b-a)^2}{12}$